12.1 Cross Sections of Solids

Name _____ Period ____ Date _____

Geometry 12.1 · Student sheet

I can…

- tell whether a solid is a polyhedron and name it.
- use Euler's Theorem to find the number of faces, vertices, or edges.
- describe the shape of a cross section of a solid.

Vocabulary · fill in the missing word

Polyhedron — a solid that is bounded by _____, called faces

Face — a polygon that forms a side of a _____

Edge — a segment where two faces of a _____ meet

Vertex — a point where three or more edges of a _____ meet

Cross section — the _____ of a plane and a solid

Part 1 · Discovery: count faces, vertices, and edges

Count the faces (F), vertices (V), and edges (E) of each solid.

Solid	F	V	E	F + V
Cube				
Triangular prism				
Square pyramid				
Pentagonal prism				

F + V is always _____ more than E.

Part 2 · Notes and examples (fill in each note, then work the example)

Prisms and pyramids: a prism has two congruent, parallel _____; a pyramid has one base, and its lateral faces meet at a _____.

Example 1

Tell whether each solid is a polyhedron. If it is, name it.

a. two parallel pentagon bases and five rectangle faces

b. a cone

c. a square base and four triangle faces that meet at a point

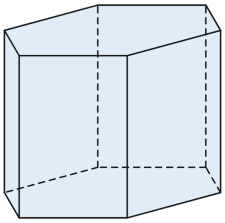
Euler's Theorem: for any polyhedron, F + V = _____

Example 2

A polyhedron has 12 faces and 30 edges. How many vertices does it have?

Example 3

Find the number of faces, vertices, and edges of the hexagonal prism. Check that Euler's Theorem holds.



Cross section: the shape formed where a _____ slices through a solid.

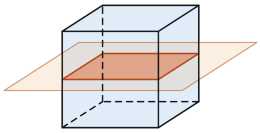
Example 4

Describe the shape of each cross section.

a. a cube cut by a plane parallel to a face (shown)

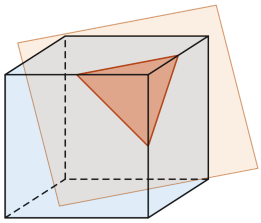
b. a cylinder cut by a vertical plane through the centers of its bases

c. a cone cut by a plane parallel to its base



Example 5

A plane cuts off one corner of a cube, as shown. What shape is the cross section?



Part 3 · Partner practice

Partner 1

Count the faces, vertices, and edges of a square pyramid. Show that Euler's Theorem holds.

Partner 2

Describe the cross section of a square pyramid cut by

a. a plane parallel to its base

b. a vertical plane through its vertex and the midpoints of two opposite base edges